

Excel Functions — Detailed Lecture Notes

JKSSB Computer Awareness

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26.1 From formulas to functions

MS Excel is built around the idea that a value can be *calculated* instead of typed. A cell that begins with the equal sign = holds a calculation, and Excel works out the result. Such a calculation is called a **formula**. A simple formula is =A1+A2, which adds the values of two cells. A formula can also use operators such as +, -, *, and / to add, subtract, multiply, and divide.

Typing every cell into a formula becomes slow and error-prone when there are many cells. Excel therefore provides **functions**: ready-made, named operations that do a common job for you. SUM adds a range of numbers, AVERAGE finds their mean, and MAX finds the largest. A function is always written *inside* a formula. The function name is followed by parentheses that contain the *arguments* — the cells or values the function should work on.

Definition (Formula). *An expression that begins with the equal sign = and calculates a value, for example =A1+A2 or =SUM(A1:A10).*

Definition (Function). *A predefined, named built-in operation, such as SUM, COUNT, or IF, that is used inside a formula to perform a particular calculation on its arguments.*

The distinction to hold is simple. A **formula** is the whole calculation. A **function** is a built-in tool that performs one part of that calculation. Every function is written as a formula, but not every formula contains a function: =A1+A2 is a formula with no function in it, while =SUM(A1:A10) is a formula that uses one function. A formula must begin with =; if the equal sign is missing, Excel treats the entry as plain text and shows it as typed.

26.2 Function syntax and arguments

Every function follows the same shape: a name, then a pair of parentheses, and inside the parentheses the arguments separated by commas. This is written as =FUNCTION(argument1, argument2). The **name** tells Excel which operation to perform. The **arguments** tell it which cells or values to use. For example, in =SUM(A1:A10) the name is SUM and the single argument is the range A1:A10.

A **range** is a block of cells written with a colon between the first and last cell. A1:A10 means the cells A1, A2, A3, and so on down to A10. A function can take more than one argument: =SUM(A1:A5, C1:C5) adds the two ranges together. The comma separates one argument from the next. Knowing the name, the parentheses, and the colon is enough to read any of the basic functions in this lesson.

Function	Example	What it returns
SUM	=SUM(A1:A10)	The total of the numbers in the range.
AVERAGE	=AVERAGE(A1:A10)	The arithmetic mean of the numbers.
COUNT	=COUNT(A1:A10)	The number of cells holding numbers.
COUNTA	=COUNTA(A1:A10)	The number of non-empty cells.
MAX	=MAX(A1:A10)	The largest number in the range.
MIN	=MIN(A1:A10)	The smallest number in the range.
IF	=IF(A1>50,"Y","N")	One of two values, chosen by the test.

26.3 SUM and AVERAGE

SUM returns the total of the numbers in its arguments. Its most common form is =SUM(A1:A10), which adds all the numbers in the range A1:A10. Cells that hold text and cells that are blank are ignored; only numbers are added. SUM accepts several ranges, so =SUM(A1:A5,B1:B5) adds both blocks in one step.

AVERAGE returns the arithmetic mean of the numbers in its arguments. =AVERAGE(A1:A10) adds the numbers and divides by how many numbers there are. The important detail is what counts in the divisor: blank cells and cells holding text are *not* counted. AVERAGE divides by the number of *numeric* cells, not by the total number of cells in the range. This is the source of many wrong answers.

Example 26.3.1. Suppose A1:A5 holds the values 10, 20, and 30, while A4 and A5 are blank. =SUM(A1:A5) returns 60. =AVERAGE(A1:A5) returns $(10 + 20 + 30)/3 = 20$, not $60/5 = 12$. Excel divides by the three cells that actually hold numbers, because the two blanks are ignored.

26.4 COUNT and COUNTA: the near-neighbour pair

COUNT and COUNTA look almost the same but count different things, and the difference is a favourite exam trap. COUNT(counts) only the cells in a range that contain *numbers*. COUNTA (the “A” stands for *all*) counts every cell that is *not empty*, whether it holds a number or text. Both functions ignore blank cells.

Definition (COUNT). A function that returns the number of cells in a range that contain numeric values. Text, blanks, and logical values are not counted.

Definition (COUNTA). A function that returns the number of cells in a range that are not empty. It counts numbers, text, and error values alike; only truly blank cells are ignored.

The contrast is easiest to see with a mixed range. Suppose A1:A10 holds six numbers, two text labels, and two blank cells. =COUNT(A1:A10) returns 6, because only the numeric cells are counted. =COUNTA(A1:A10) returns 8, because the six numbers and the two text labels are all non-empty. Both functions ignore the two blanks. A related function, COUNTBLANK, does the opposite and counts the empty cells, returning 2 for the same range. For the exam, the pair to memorise is: **COUNT = numbers only; COUNTA = all non-empty cells.**

26.5 MAX and MIN

MAX returns the largest number in a range, and MIN returns the smallest. Their forms are =MAX(C1:C10) and, for the smallest, =MIN(C1:C10). Like SUM, AVERAGE, and COUNT, they

work on numbers and ignore text and blank cells. A useful way to remember them is that **MAX** is the maximum and **MIN** is the minimum; do not confuse **MIN** with the word “minimum cell”, which is not a function at all.

Example 26.5.1. *A range holds the marks 42, 67, 55, 91, and 38. =MAX(C1:C5) returns 91, the top mark. =MIN(C1:C5) returns 38, the lowest mark. =AVERAGE(C1:C5) returns 58.6, the mean of the five numbers.*

26.6 The IF function and nested IF

IF lets a cell show one of two results depending on whether a condition is true or false. Its syntax is `=IF(logical_test, value_if_true, value_if_false)`. The first argument is a *logical test* that can be answered only with true or false, such as `A1>50`. The second argument is the value or text to return when the test is true, and the third is the value to return when it is false. Thus `=IF(A1>50,"Pass","Fail")` shows **Pass** when A1 is more than 50 and **Fail** otherwise.

A **nested IF** places a second IF inside the value-if-false part of the first, so more than one condition can be tested. A typical form is `=IF(A1>50,"High",IF(A1>25,"Medium","Low"))`. Excel checks the conditions in the order they are written, and the first test that is true decides the result.

Example 26.6.1. *Take the formula =IF(A1>50,"High",IF(A1>25,"Medium","Low")) and let A1 be blank. In a numeric comparison Excel reads a blank cell as 0. Since 0 is neither greater than 50 nor greater than 25, both tests are false and the formula returns the final value, Low. This is the behaviour the Junior Assistant 2026 paper examined (KB PYQ-14, JA 2026 Q74).*

When the input is *non-numeric* text, the behaviour depends on what the formula does with it. A comparison of text against a number is still answered by Excel, but if the formula tries to *calculate* with text — for example, `=A1*2` where A1 holds a word — Excel cannot do the arithmetic and returns the error value **#VALUE!**. The safe exam reading is: a blank is treated as 0 in a numeric test, and text that is arithmetic-only produces **#VALUE!**.

26.7 A worked sheet, function by function

A small sheet makes the behaviour of every function concrete. Column A holds five student names — Anita, Bilal, Chandan, Divya, and Esha — and column B holds the marks 40, 55, 70, and 60, with Esha’s mark (B5) left blank because she has not been marked yet. Reading the sheet function by function shows what each one does with the blank.

Formula	Result	Why
=SUM(B1:B5)	225	Adds 40, 55, 70, 60; the blank is ignored.
=AVERAGE(B1:B5)	56.25	Divides 225 by 4 numeric cells, not by 5.
=COUNT(B1:B5)	4	Counts the four cells that hold numbers.
=COUNTA(B1:B5)	4	Counts the four non-empty marks; B5 is blank.
=MAX(B1:B5)	70	The largest of the four marks.
=MIN(B1:B5)	40	The smallest of the four marks.
=IF(B5>=40,"Pass","Fail")	Fail	B5 is blank, read as 0, so the test fails.

The last row links the sheet back to the nested-IF behaviour: a blank cell is read as zero in a numeric test, so Esha is shown as not yet passing. Note also that **COUNTA** on the names column, **=COUNTA(A1:A5)**, would return 5, because all five names are present even though only four marks are. This is exactly how the two count functions are used together to spot missing data.

26.8 Other basic functions worth knowing

Beyond the core set, a few basic functions appear in Office-suite questions. **COUNTIF** counts the cells that meet one condition, and **SUMIF** adds the cells that meet one condition; both extend **COUNT** and **SUM**. On the statistics side, **MEDIAN** returns the middle value of the numbers, while **MAX** and **MIN** return the ends of the range. **ROUND** rounds a number to a stated number of digits. **LEN** counts the characters in a text value. **TODAY** returns the current date, and **NOW** returns the current date and time.

Function	What it returns
COUNTIF	The number of cells meeting one condition.
SUMIF	The total of the cells meeting one condition.
MEDIAN	The middle value of the numbers.
ROUND	A number rounded to a set number of digits.
LEN	The number of characters in a text value.
TODAY	The current date.

At JKSSB depth, the exam is satisfied by knowing what each function returns and that every one of them is written inside a formula that starts with **=**.

26.9 AutoSum

AutoSum is a button, not a function, that quickly inserts a **SUM** formula. When the active cell is placed below or beside a column or row of numbers, AutoSum looks at the adjacent numbers and writes a **=SUM(...)** formula over the range it selects. Pressing the button therefore saves the user

from typing the range by hand. The keyboard shortcut for AutoSum is **Alt + =**. Its drop-down arrow also offers **Average**, **Count Numbers**, **Max**, and **Min**, so the same button can insert any of the core functions, not only SUM.

Definition (AutoSum). *An Excel button that automatically inserts a SUM formula for the range next to the active cell; the shortcut is Alt + =.*

26.10 Combining functions in one formula

Functions can be nested inside one another, so the result of one function becomes the argument of another. For example, `=ROUND(AVERAGE(A1:A10),1)` first finds the mean of A1:A10 and then rounds it to one decimal place. In the same way, `=IF(COUNT(B1:B5)=0,0,AVERAGE(B1:B5))` shows 0 when the range contains no numbers at all, instead of producing an error, and otherwise shows the mean of the numbers present. Nesting is read from the inside out: the innermost function is calculated first, and its result is passed outwards to the next function.

Example 26.10.1. *Consider `=IF(COUNT(C1:C6)=0,"No data",AVERAGE(C1:C6))`. If C1:C6 holds only text or blanks, COUNT returns 0, the test is true, and the cell shows No data. If the range holds numbers, the test is false and the cell shows their average. This pattern is common on real marks sheets, where a column may be empty early in the term.*

The arguments of a function can also be single cells, several separate cells, whole ranges, or even constant values such as `=SUM(A1, 5, B3:B7)`. A reference such as A1 can be relative or absolute (`$A$1`); that choice does not change what the function does, only how the reference behaves when the formula is copied.

26.11 Error values

When a formula cannot be calculated, Excel displays an *error value* beginning with the hash sign #. Each error points to a different kind of problem, and recognising them is a common direct-recall question.

Error value	What it means
#DIV/0!	A number is divided by zero or by a blank cell.
#VALUE!	The wrong type of argument is used — for example, text where a number is needed.
#NAME?	Excel does not recognise a name in the formula, often a misspelt function name.
#REF!	A cell reference is not valid, usually because the referenced cell was deleted.
#N/A	The value is not available to the formula, as when a lookup finds no match.
#NUM!	A number in the formula is not valid for the operation.
#NULL!	A space (intersection) between two ranges that do not intersect.

Two clarifications prevent confusion. First, the string ##### that sometimes appears in a cell is *not* an error value; it only means the column is too narrow to display the number, and it disappears

when the column is widened. Second, **#DIV/0!** also appears when the divisor is a blank cell, because a blank is read as zero in the division. Reading each error as a clue to what Excel could not do is the fastest way to answer these items.

Example 26.11.1. *The error value points at the mistake that produced it. A formula `=A1/0`, or `=A1/B1` where *B1* is blank, shows **#DIV/0!**. A formula `=A1+B1` where *A1* holds text shows **#VALUE!**. A misspelt function such as `=SUMM(A1:A5)` shows **#NAME?**. Deleting a cell that a formula refers to turns the reference into **#REF!**. A lookup that finds no match leaves the result **#N/A**, and a mathematically invalid argument such as `=SQRT(-1)` gives **#NUM!**.*

26.12 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. A **formula** starts with `=`; a **function** is a named built-in operation used inside a formula. Without the equal sign, the entry is text (TRAP-30).
2. **COUNT** counts numbers only; **COUNTA** counts all non-empty cells. Do not swap this near-neighbour pair.
3. **AVERAGE** divides by the number of numeric cells, not by the total number of cells.
4. A **blank** cell is read as zero in a numeric test, so a blank fails a “greater than” condition; text used in arithmetic gives **#VALUE!** (KB PYQ-14).
5. **AutoSum** inserts `=SUM(...)` and its shortcut is **Alt + =**, not a Ctrl combination.

Example 26.12.1. *An item asks which function counts a range that mixes numbers and labels. Read the function name carefully: **COUNT** returns only the numeric cells, while **COUNTA** returns every non-empty cell. If the stem asks for the count of entries of any kind, the answer is **COUNTA**; if it asks for the count of numbers, the answer is **COUNT**. A second item may ask which error appears when a cell is divided by a blank; the blank counts as zero, so the error is **#DIV/0!**.*

Exercises

1. Explain the difference between a formula and a function in MS Excel.
2. A range *A1:A8* contains 5 numbers, 2 text labels, and 1 blank cell. State the values returned by `=COUNT(A1:A8)` and `=COUNTA(A1:A8)`.
3. Write the general syntax of the **IF** function and explain each of its three arguments.
4. A cell holds `=IF(A1>10,"Yes",IF(A1>0,"Small","No"))` and *A1* is blank. What is shown, and why?
5. Match each error value with its cause: **#DIV/0!**, **#VALUE!**, **#NAME?**, **#REF!**.

Solutions

1. A formula is any calculation that begins with =, such as =A1+A2. A function is a predefined, named operation such as SUM or IF that is used inside a formula. Every function is written as a formula, but a formula need not contain a function.
2. =COUNT(A1:A8) returns 5, because only the numeric cells are counted. =COUNTA(A1:A8) returns 7, because the 5 numbers and the 2 text labels are all non-empty; both ignore the blank.
3. =IF(logical_test, value_if_true, value_if_false). The first argument is a test answered true or false; the second is the value returned when it is true; the third is the value returned when it is false.
4. The formula shows No. A blank is read as zero in a numeric test, so A1 is neither greater than 10 nor greater than 0, both tests are false, and the final value No is returned.
5. #DIV/0! — division by zero or by a blank cell; #VALUE! — the wrong type of argument, such as text where a number is needed; #NAME? — an unrecognised name, often a misspelt function; #REF! — an invalid cell reference after the referred cell was deleted.